Digital Systems and Computer Organization

MIDSEM PRACTICE PAPER | 30 marks | Suggested time: 90 minutes

Modules 1-4; module 5 through memory operations

Digital-system questions selected to match your scope; Teams module 4 lists flip-flops, registers, ripple/ring counters, synchronous counters and Verilog. The module-5 PPT was not visible. Q10 on byte/word addressing is a provisional match to the memory-operations cutoff. No cache, pipelining or assembly-programming questions included.

Answer all questions. Show working and draw labelled diagrams where appropriate. Marks and duration are a practice format, not a confirmed university blueprint.

Question 1 [3 marks]

Represent the following numbers in 5-bit 2's complement form, perform the operation mentioned and state whether overflow occurs or not in each case. i) Add +9 and +12 ii) Add -8 and -9 Subtract -5 from +10

Source: DSCO-M | original Q1B

Question 2 [3 marks]

Simplify F(A,B,C,D) = ΠM(0,3,5,6,8,10), with don’t-care terms d(2,13,14), and implement it using NOR gates only.

Source: DSCO-R | original Q1B

Question 3 [3 marks]

Design a 3-bit magnitude comparator for unsigned binary numbers A = $a_2a_1a_0$ and B = $b_2b_1b_0$. Derive the final Boolean expressions and draw the equivalent logic circuit.

Source: DSCO-R | original Q1C

Question 4 [3 marks]

Implement a three-input XOR gate using 4-to-1 multiplexer with a neat diagram.

Source: DSCO-M | original Q1C

Question 5 [2 marks]

Draw PLA implementations for the following Boolean expressions:
(i) F = ABC′ + ABC + A′BC + AB′C + AB′C′.
(ii) F1 = A′B + BC′ and F2 = AB′ + A′C.

Source: DSCO-R | original Q5C

Question 6 [5 marks]

Design a sequential circuit with two JK flip-flops A and B and two inputs E and F. If E = 0, the circuit remains in the same state regardless of the value of F. When E = 1 and F = 1, the circuit goes through the state transitions from 00 to 01, to 10, to 11, back to 00, and repeats. When E = 1 and F = 0, the circuit goes through the state transitions from 00 to 11, to 10, to 01, back to 00, and repeats.

Source: DSCO-M | original Q2A

Question 7 [3 marks]

Design a 3-bit synchronous binary up-counter using T flip-flops. Draw the logic diagram.

Source: DSCO-M | original Q2B

Question 8 [3 marks]

Illustrate a 4-bit asynchronous ripple counter with a neat diagram using JK Flip flops and explain the working of the same.

Source: DSCO-R | original Q2B

Question 9 [2 marks]

Write behavioral Verilog code that represents an eight-bit ring counter.

Source: DSCO-R | original Q2C

Question 10 [3 marks]

Explain the concept of memory addresses in an Instruction Set Architecture. Also, discuss the difference between byte addressing and word addressing.

Source: DSCO-R | original Q3B

Sources and selection notes

Working interpretation: NSM means end-semester. This has not yet been confirmed. Questions come only from the local semester-3 CSE papers, using the exact current subject titles. R = regular; M = makeup. Wording is lightly edited, tables transcribed, original question references retained. No new question scenarios were invented.

DSCO-M: 2025-26 Makeup - CSS_2104-Digital Systems and Computer Organization

Local source: /Users/shritaake/Desktop/code/paper/public/authoritative/MIT 2021-26/MIT 2025-26 QPs 3rd March 2026/December 2025 and January 2026 (Makeup)/MIT Sem 1 to Sem 7 QPs/Semester 3/CSS_2104-Digital Systems and Computer Organization.pdf

DSCO-R: 2025-26 Regular - CSS_2104-Digital Systems and Computer Organization

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